

A: Provide a question that can be answered using structured query language (SQL) to acquire data from *both* the original database and the add-on CSV file data. The question should require data from *both* data sources.

How many customers have used the various internet services in each state?

A1: Identify which data from the original data set and the add-on CSV file are needed to answer the question including all tables, columns, and data types.

To be able to answer the research question identify the number of customers that have ever utilized the internet services in all the states, I will extract the state names from the location table and count the number of customers in the services who have internet services and exclude those customers without internet services. Since there is no relationship between the location table and the add on csv data in the services table, I will utilize the customers table as a linking table.

Original data set

table name	column name	data type
location	state	text

Add-on csv

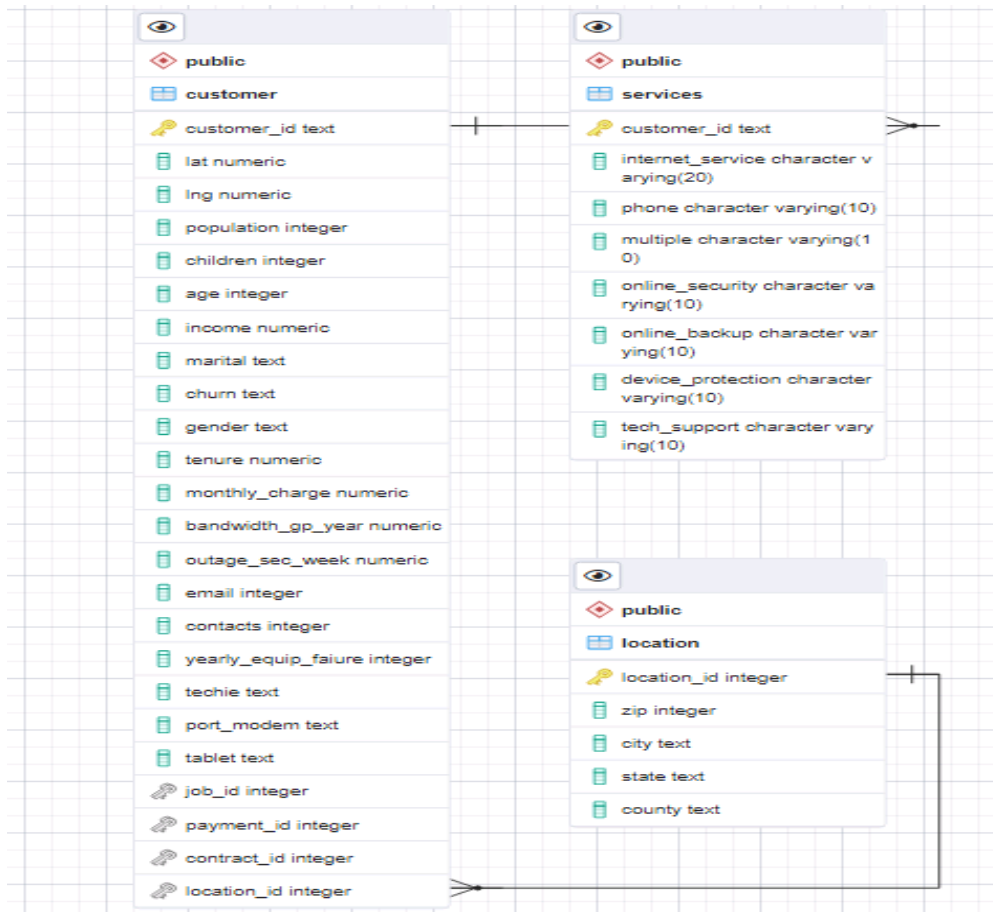
table name	column name	data type	exclude
services	Customer_id	text	
services	internet_service	character varying (20)	internet_services = none

Linking table

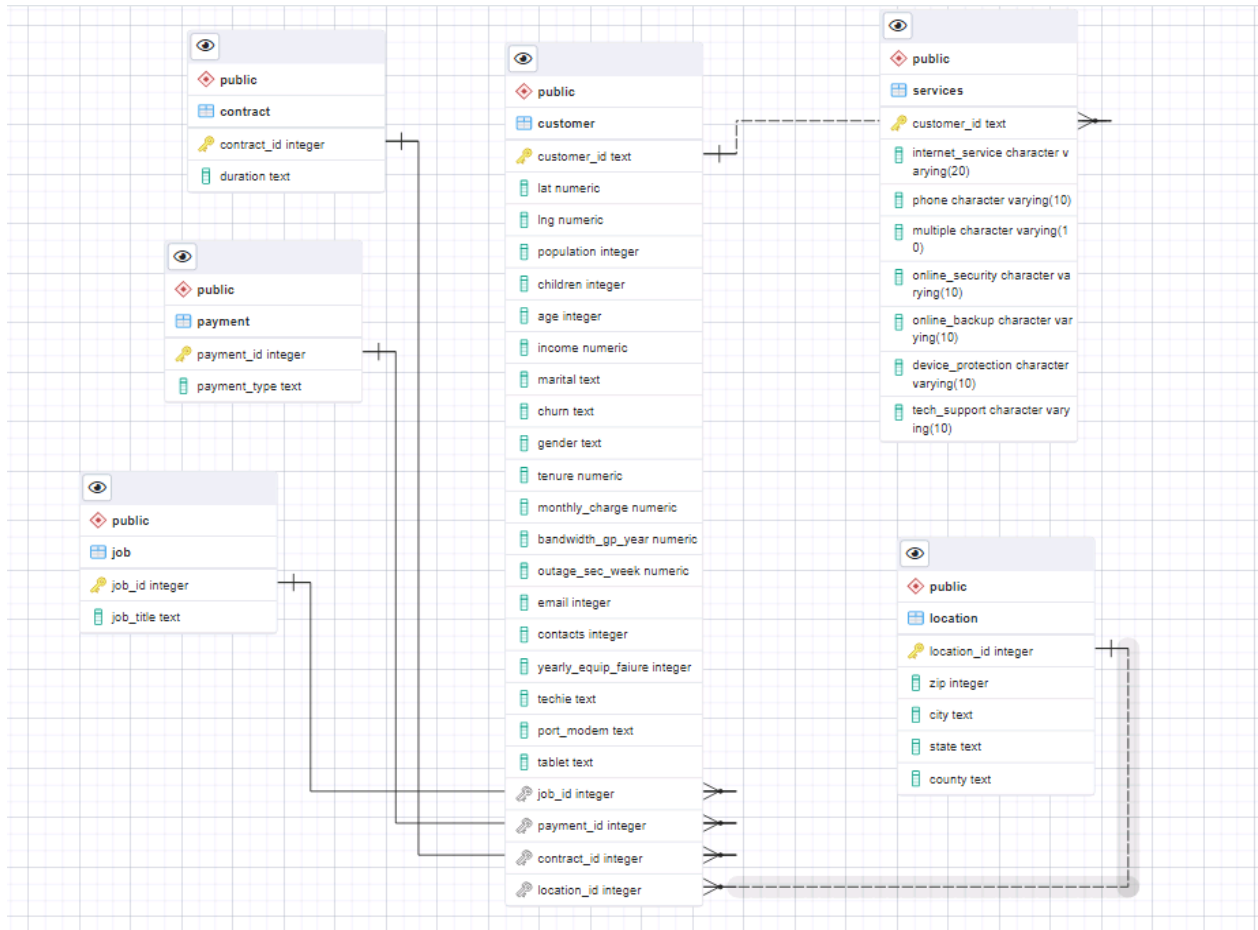
table name	column name	data type
customer	customer_id	text

B: Create an entity relationship diagram (ERD) for the add-on CSV file and any other tables and columns used to answer the question from part A by evaluating the data contained in the file and identifying the m: n relationships and relational constraints.

ERD diagram between the add-on csv (under service table), customers table and Location table



ERD for the Churn database



Malik,U.,Goldwasser,M,& Johnston,B. (2019) Sql for data Analysis pg 72 state that foreign keys can be added as a column constraint in order to improve the integrity of the data by making sure that the foreign key never contains a value that cannot be found in the referenced table.

The table customer references the location table using the foreign key constraint. The relationship is many to one relationship whereby many customers can reside in the same location. The relationship between the two tables is through the `location_id` field in customer table and `location_id` field in location table.

The table services reference the customer table using the foreign key constraint. The relationship is one to many relationships whereby one customer can have many services. The

relationship between the two tables is through the customer_id field in customer table and customer_id field in services table.

B1: Write SQL code, in text format, that creates a table based on the ERD and specifies the columns and relevant keys.

```
CREATE TABLE services (  
    customer_id Text NOT NULL PRIMARY KEY,  
    internet_service Varchar (20),  
    phone varchar (10),  
    multiple varchar (10),  
    online_security varchar (10),  
    online_backup varchar (10),  
    device_protection varchar (10),  
    tech_support varchar (10)  
  
);  
  
--Creating referential integrity  
  
ALTER TABLE services  
  
ADD CONSTRAINT services_fk_customer
```

```
FOREIGN KEY (customer_id)
REFERENCES customer(customer_id)
ON DELETE CASCADE;
```

B2: Write SQL code, in text format, that loads the data from one of the add-on CSV files into the table created in part B1.

```
COPY services FROM 'C:/LabFiles/services.csv' DELIMITER ',' CSV HEADER
QUOTE '';
```

C: Write a SQL statement or statements in text format for a query or queries that answer the question from part A.

```
SELECT state,
        COUNT (DISTINCT s.customer_id) AS number_of_customers
FROM services s
INNER JOIN customer c ON s.customer_id= c.customer_id
INNER JOIN location l ON c.location_id= l.location_id
WHERE s.internet_service != 'none'
GROUP BY state ;
```

C1: Provide a data file or files that capture the results from the query or queries.

Attached filename: data-1707590696079.csv



D: Identify the specific time period for how often the add-on file should be acquired and refreshed in the database for the data to remain relevant to the business and the question from part A.

Daily

D1: Explain why the time period identified in part D is relevant to the business needs.

The cadence of daily for the acquisition of the add-on file and the refresh in the database is to allow for delta loads.

Mohit,M.,Kishore D, & Meenakshi P.S. (2022,January 03) state that Delta data load is a technique for loading incremental changes after an initial full load to a data. It is an alternative to running a change data capture (CDC) process which replicates data changes as they happen. A delta load implies that the entire data of a relational database table is not repeatedly extracted, but only the new data that has been added to a table since the last load. With delta load, you can process only data that needs to be processed, either

new data or changed data. This makes the process efficient, reducing costs and saving time.

Based on the nature of a telecom industries, new customers are acquired on a daily basis, existing customers can request for additional services and other customers disconnect services on a daily basis. To be able to perform an accurate data analysis and answer the business question of the number of customers utilizing our internet services per state, then the data need to be refreshed on a daily basis.

E: Provide a Panopto video recording that includes the presenter and a vocalized demonstration showing all code used, the code being executed, and the results of all code used in the task.

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=577959ca-801d-44e2-b72f-b112014a76f2>

E1: Include a vocalized demonstration within the Panopto video recording provided in part E that describes the programs used to complete the task.

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a481d7cd-bd6f-4e0a-a7eb-b112015207ad>

F: Acknowledge web sources used to acquire data or segments of third-party code to support the application. Be sure the web sources are reliable.

There were no third-party sources used to acquire data or any segment of third-party code.

G: Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

References

Malik,U.,Goldwasser,M,& Johnston,B. (2019) Sql for data Analysis. Birmingham, UK: Packt Publishing Ltd.

Mohit,M.,Kishore D, & Meenakshi P.S. (2022,January 03),Perform delta data loads to data lakes using AWS DMS. <https://aws.amazon.com/blogs/database/perform-delta-data-loads-to-data-lakes-using-aws-dms>